

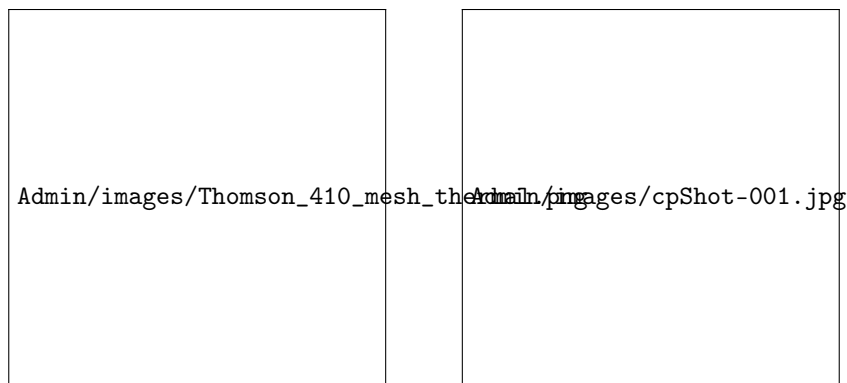


Figure 1: An energy distribution modeled with circle packing.

2013 HBCU-UP Mini Grants Program Request for Proposals

1 Purpose and specific hypotheses/research questions

Circle packings are configurations of circles with specified patterns of tangency. They naturally express a conformal geometry which has allowed the field of circle packing to be developed into a rich theory of discrete analytic functions, see [?, ?]. The discretization of complex analysis via computable circles has application advantages. For example, the NSF has supported research for using circle packings in brain imaging, see [?, ?, ?].

Using circle packings Dr. Ken Stephenson has constructed discrete models of the Thomson problem. A longstanding problem in physics with applications in biology, the Thomson problem asks how to find the minimum energy distribution of a finite number of electrons on the surface of the sphere. By applying a Delaunay triangulation to any distribution, an approximate spherical circle packing model can be found, see figure 1. Furthermore, combinatorial manipulations can be used to minimize the total energy. It is hoped that exploring these manipulations will provide insights into the problem.

1.1 What will be done

We propose that Johnson C. Smith undergraduate students be tasked with exploring the Thomson problem via circle packings. Students should be able to construct discrete models and measure their approximation of the classical problem. Through combinatorial manipulation, they will experiment with distributions and seek to make conjectures about the problem.

1.2 Estimated start-up and completion dates

It is expected that work can begin by February 2013 and continue until May 2013. Because of the variety of possible experiments, there is potential that this work may develop into a larger project.

1.3 How

See Section 2 below.

1.4 By Whom

The necessary code for implementing and manipulating the spherical packing models will be written by Dr. Stephenson and Dr. Ashe. Thus the student(s) would need little prerequisite programming or mathematical knowledge, only a willingness to learn and the patience to conduct the experiments.

JCSU student Daniel Villanueva and Merischia Griffin have been recommended for this project. It is an excellent opportunity to be introduced to programming and high level mathematics.

2 Procedures for evaluation

Circle packings can be manipulated by Java™ code or directly through **CirclePack**. A configuration of circles (location and radii) can then be computed with **CirclePack**. Energy points are identified with circle vertices, and energy is then easily calculated with software. Although there has been little study of the Thomson problem using circle packings, the problem has been thoroughly researched. Thus there is an abundance of available data for the derivation of discrete models.

The creation and analysis of these experiments are done strictly on computers. The following materials will be required:

- **CirclePack**; creates and manipulates circle packings. This software is open source and freely available.
- Eclipse; a multi-language software development environment. This software is open source and freely available.
- Java™; **CirclePack**'s programming language. This software is freely available.
- **MATLAB®**; a numerical computing and analysis program.
- Suitable computer hardware for using the above software.

3 Budget

The required software **CirclePack**, Java™, and Eclipse are freely available through public licenses. Student and faculty licenses for **MATLAB®** should be available through existing JCSU resources. Thus costs will be limited mostly to compensation and travel expenses. An estimate is as follows:

- \$1000: Stipend for Dr. Ashe.
- \$1000: Stipends for two students (\$500 each).
- \$1482: Travel allowance for students and Dr. Ashe.
- \$518: Hosting Dr. Stephenson at JCSU from February 10, 2013 to February 12, 2013.
 - \$238 (\$119 per night): Lodging at the Mosaic Village for Dr. Stephenson
 - \$175.04 (233.8 miles at \$0.375 per mile): Dr. Stephenson's mileage cost from his home in Knoxville, TN to JCSU.
 - \$105 (\$35 per day): per diem for Dr. Stephenson.
- \$300: Computer hardware (this cost may be reduced or eliminated with corporation with the IT department).
 - An additional monitor.
 - Graphics card capable of supporting dual monitors and handling 3D modeling.
- \$100: Presentation materials.

Additional expenses to be paid for by HBCUP:

- \$1000: Stipend for visiting scientist Dr. Stephenson. This cost also pays Dr. Stephenson's honorarium for his visit on February 10, 2013 to February 12, 2013.

4 Time required

There is a very large number of possible experiments, and anything discovered would be of value. Thus students will be compensated at the standard rate until their stipends are exhausted.

Due to the nature of programming and experimental mathematics, the time required by the faculty is difficult to estimate. In addition to programming, the faculty would need to advise students, create images, and attach mathematical rigor to the students' experimental results. An estimate of 40+ hours for the faculty advisors might be conservative. Thus the stipend compensation will most certainly cost less than that of a programmer/mathematician being paid at the standard rate.

5 Ethical Issues

There are no known ethical issues particular to this project.

6 Benefits of Research

Benefits to the study of mathematics are outlined in section 1. The students will simultaneously gain experience in programming, algorithms, optimization, and pure mathematics. Results might be publishable, garner note in a high level math publication, or be the cornerstone of a senior project. Additionally, the work would enhance a résumé, graduate school application, or portfolio.

7 Plan to Share and Disseminate

This visual nature of circle packings are excellent for capturing the attention of mathematicians and lay people alike. Posters and the junior seminar would be appropriate formats to display and discuss early results. Further along, regional undergraduate math conferences and math journals would be the next step. Dr. Stephenson will visit JCSU, work with students, and speak at a STEM College lyceum on February 11, 2013.